

ADRIAN SHKUMBOV

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EDUCATION

University of Toronto
Bachelor of Science in *Engineering Science*

Graduation Date: *Spring 2026*
Specialization: *Robotics Engineering*

SKILLS

Mechanical: SOLIDWORKS, Fusion360, AutoCAD, 3D Printing, Laser Cutting, Basic Machining, Power Tools
Electrical/Embedded: Micro-controllers, Embedded Platforms, KiCAD, EaglePCB, Oscilloscope, Soldering
Software: Python (NumPy, OpenCV, PyTorch), C, ROS2, MATLAB, Simulink, $\text{\LaTeX}$

EXPERIENCE

Mechatronics Design Intern – Martinrea International, *Vaughan* 05/2024 – 09/2025

- Designed and deployed custom computer vision systems (Python/OpenCV) for internal manufacturing lines globally.
- Started and led an R&D project to develop a novel robotic vision system for surface inspection and data collection.
- Designed and manufactured prototypes of an electro-mechanical camera protection device for harsh environments.
- Spent extended periods abroad leading high-profile company endeavours regarding technical assessment and software improvements on large-scale vision systems in response to urgent production concerns.

Engineering Research Intern – University of Strathclyde, *Glasgow* 05/2023 – 08/2023

- Worked under Prof. Enrico Tubaldi on a project to devise low-cost effective systems for the monitoring of bridge scour.
- Developed MATLAB scripts to fuse GPS and depth data from sonars to visualize the state of riverbeds and scour holes.
- Carried out sonar mapping tests in the field and submitted technical reports.

Propulsion Sub-team Member – University of Toronto Hyperloop Team, *Toronto* 09/2022 – 03/2023

- Led the electrical work for the creation of a small-scale Linear Halbach Drive.
 - Developed embedded C code for motor control and devised bench tests to drive progress.
 - Used SOLIDWORKS to model and 3D print specialized housings for electrical components and sensors.
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PROJECTS

Torque Vectoring System 10/2025 – 12/2025

- Designed (in SOLIDWORKS) and built a small-scale sensed electric vehicle with a rear BLDC electronic differential.
- Co-developed micro-controller code to detect traction loss from IMU & encoder data and perform torque redistribution.
- Devised a Kalman filter to incorporate vehicle kinematic model with sensor data for more robust operation.

Unicycle Robot 02/2024 – 04/2024

- Designed and built a control moment gyroscope based unicycle robot using 3D printing and laser cutting.
- Selected actuators, sensors, and micro-controller for desired functionality within budget constraints.
- Implemented (in C) PID control loops, digital filters, and interrupt service routines on micro-controller.

Wearable Real-Time EMG Visualizer 03/2024 – 04/2024

- Designed and soldered a dual-channel EMG sensing PCB using op-amps for amplification, filtering, and rectification.
- Integrated sensing PCB with micro-controller and addressable LED strips to create a shirt displaying live bicep exertion.